Exp:04

Implement Packet Sniffing In Wireshark

Aim:-

To capture and analyze various network protocol packets (like TCP, UDP, ARP, DNS, HTTP, ICMP, DHCP) using Wireshark and apply display filters to inspect them.

Software Used:-

Wireshark Network Protocol Analyzer

Procedure:-

General steps (for all captures):-

1. open Wireshark
2. select local area connection
3. Go to Captures -> options
4. set: stop capture automatically after 100. packets.
5. click start to begin capturing
6. Use filter bar to type specific protocols like tcp, arp, etc.
7. Use statistics -> Flow Graph.
8. Click file -> Save to store the captured packets.

Tasks & outputs:

1. Filter : Display TCP/UDP packets and Flow Graph.

* Filter used : tcp or udp
* output : Displayed only TCP/UDP packets.
* Flow Graph : Viewable from statistics → flow Graph.

2. Filter : Display only ARP packets

* Filter used : arp
* output : Displayed only Address Resolution protocol packets.

3. Filter : Display DNS packets and flow Graph.

* Filter used : dns
* output : Displayed only DNS quires and responses.
* flow Graph : Shows DNS request - response communication.

4. Filter : Display only HTTP packets

* Filter used : http

★ output: captured HTTP traffic (browser or server communication).

5. Filter: Display IP/ICMP packets

★ Filter used: icmp or ip

★ output: shows ICMP packets (used in ping operations) and IP packet info.

6. Filter: Display only DHCP packets.

★ Filter used: bootp (used for DHCP)

★ output: Displays DHCP Discover, offer, Request, Acknowledge packets.

Result:-

Successfully captured and filtered different protocol packets using Wireshark and understood how to inspect and analyse them with tools like filters and flow graphs.